Term Test Solutions

EEE 0713 3237 (EEE 337): Power System II · SUST

Our batch: TT-1, TT-2 and TT-3, solved the way to write them in the exam

Term test	Question	Marks	Final answer
TT-1	Q1 Inductance of a double-circuit line	5	L = 0.431 mH/km/phase
	Q2 Transposition + capacitance, unsymmetrical spacing	5	$C_n = 2\pi k / \ln(D_{eq}/r)$ F/m
TT-2	Q1 Sag with wind pressure	4	Slant sag 20.08 m, vertical sag 12.0 m
	Q2 Permissible current loading of a cable	3	I = 580.5 A
	Q3 Varley loop test	3	Fault at 37.5 km
TT-3	Q1 Sudden change in mechanical input (P–δ curve)	4	Equal-area explanation
	Q2 String of three disc insulators	6	K = 0.375, V_L = 64.3 kV, η = 68.3 %

Source: 04_Term_Tests_Questions/ (Ours_TT-1.pdf, Ours_TT-2.jpeg, Ours_TT-3.jpeg). Symbols: ln = natural log, log = $\log_{10}$, $r' = 0.7788\ r$, $k = \varepsilon_0\varepsilon_r$, $\varepsilon_0 = 8.854\times10^{-12}$ F/m. Book references: K&N = Kothari & Nagrath, Modern Power System Analysis; Mehta = V.K. Mehta, Principles of Power System.

Exam-writing pattern used throughout: Given → Figure → Formula → Substitution → Boxed answer (with unit). Write each step on a new line exactly as shown; the examiner gives step marks.

TERM TEST 1

Total marks: 10 · Time: 30 min

Question 1

Q1. A double-circuit three-phase line is shown in Figure 1. The conductors a, a′; b, b′; c, c′ belong to the same phase, respectively. The radius of each conductor is 1.9 cm. Find the inductance of the double-circuit line in mH/km/phase. **[5]**



Figure 1: Conductor arrangement (all in one horizontal plane, 1 m apart)

Solution:

Given: radius r = 1.9 cm = 0.019 m; adjacent spacing = 1 m; a–a′, b–b′, c–c′ are the same phase. The line is assumed to be **fully transposed**, so the GMD method can be used.

Taking a as the origin, the positions are: a = 0, b = 1, c = 2, a′ = 3, b′ = 4, c′ = 5 m.

Step 1: Self GMD (GMR) of each phase

$r' = 0.7788 \times r = 0.7788 \times 0.019 = 0.0148$ m

$D_{sa} = \sqrt{(r' \cdot D_{aa'})} = \sqrt{(0.0148 \times 3)} = 0.2107$ m

Similarly $D_{bb'} = D_{cc'} = 3$ m, so $D_{sb} = D_{sc} = 0.2107$ m

$D_s = \sqrt[3]{(D_{sa}\, D_{sb}\, D_{sc})} = 0.2107$ m

Step 2: Mutual GMD between phases

$D_{ab} = \sqrt[4]{(D_{ab} \cdot D_{ab'} \cdot D_{a'b} \cdot D_{a'b'})} = \sqrt[4]{(1 \times 4 \times 2 \times 1)} = \sqrt[4]{8} = 1.6818$ m

$D_{bc} = \sqrt[4]{(D_{bc} \cdot D_{bc'} \cdot D_{b'c} \cdot D_{b'c'})} = \sqrt[4]{(1 \times 4 \times 2 \times 1)} = \sqrt[4]{8} = 1.6818$ m

$D_{ca} = \sqrt[4]{(D_{ca} \cdot D_{ca'} \cdot D_{c'a} \cdot D_{c'a'})} = \sqrt[4]{(2 \times 1 \times 5 \times 2)} = \sqrt[4]{20} = 2.1147$ m

$D_{eq} = \sqrt[3]{(D_{ab} \cdot D_{bc} \cdot D_{ca})} = \sqrt[3]{(1.6818 \times 1.6818 \times 2.1147)} = \sqrt[3]{5.9814} = 1.8152$ m

Step 3: Inductance per phase

$L = 2 \times 10^{-7} \ln(D_{eq} / D_s)$ H/m

$= 2 \times 10^{-7} \ln(1.8152 / 0.2107)$

$= 2 \times 10^{-7} \ln(8.6156) = 2 \times 10^{-7} \times 2.1536$

$= 4.307 \times 10^{-7}$ H/m

$= 4.307 \times 10^{-7} \times 10^{3} \times 10^{3}$ mH/km

∴ Inductance of the double-circuit line, L = 0.431 mH/km/phase

Reference: same type as K&N Ch-2, Problem 2.12, p.94 (the book uses r = 1.5 cm); method K&N Art 2.9 (double-circuit lines).

Question 2

Q2. Define transposition of a transmission line. Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing. **[5]**

Solution:

Transposition: When the conductors of a three-phase line are not equally spaced, the flux linkages and the charges of the three phases are different, so the inductance and capacitance of each phase are unequal. **Transposition means exchanging the positions of the three conductors at regular intervals along the line, so that each conductor occupies each of the three positions for one-third of the total length** (Fig. 2). A complete set of three exchanges is called a transposition cycle.

Why it is used: (i) it makes the average inductance and capacitance of all three phases equal, so the line becomes balanced and the voltage drops are equal; (ii) it reduces the interference (induced voltage) in nearby telephone/communication lines.

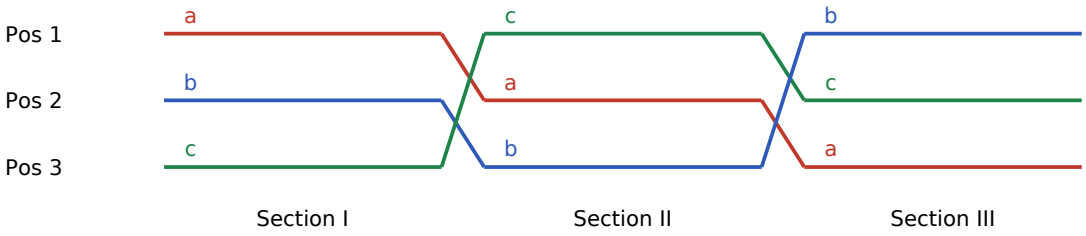
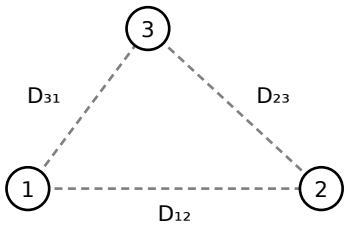


Fig. 2: One transposition cycle (each section = 1/3 of the line length)

Derivation (capacitance, unsymmetrical spacing):



q_a + q_b + q_c = 0

Fig. 3: Three conductors (radius r) at positions 1, 2, 3 with unequal spacing

Let the charges per metre on conductors a, b, c be q_a, q_b, q_c (assumed equal in every section of the transposition cycle). The potential difference between two conductors due to a group of charges is $V_{ab} = (1/2\pi k)\ \Sigma\ q_i \ln(D_{ib}/D_{ia})$.

Section I (a in position 1, b in 2, c in 3):

$$V_{ab(I)} = (1/2\pi k)\ [\ q_a \ln(D_{12}/r) + q_b \ln(r/D_{12}) + q_c \ln(D_{23}/D_{31})\ ] \quad \ldots\ldots(1)$$

Section II (a in 2, b in 3, c in 1):

$$V_{ab(II)} = (1/2\pi k)\ [\ q_a \ln(D_{23}/r) + q_b \ln(r/D_{23}) + q_c \ln(D_{31}/D_{12})\ ] \quad \ldots\ldots(2)$$

Section III (a in 3, b in 1, c in 2):

$$V_{ab(III)} = (1/2\pi k)\ [\ q_a \ln(D_{31}/r) + q_b \ln(r/D_{31}) + q_c \ln(D_{12}/D_{23})\ ] \quad \ldots\ldots(3)$$

Average V_{ab} = (1/3)[(1) + (2) + (3)]. The q_c terms add to $\ln(1) = 0$:

$$V_{ab} = (1/2\pi k)\ [\ q_a \ln(D_{eq}/r) + q_b \ln(r/D_{eq})\ ], \text{ where } D_{eq} = \sqrt[3]{(D_{12}\ D_{23}\ D_{31})} \quad \ldots\ldots(4)$$

In the same way:

$$V_{ac} = (1/2\pi k)\ [\ q_a \ln(D_{eq}/r) + q_c \ln(r/D_{eq})\ ] \quad \ldots\ldots(5)$$

Adding (4) and (5) and using $q_b + q_c = -q_a$:

$$V_{ab} + V_{ac} = (1/2\pi k)\ [\ 2q_a \ln(D_{eq}/r) - (q_b + q_c) \ln(D_{eq}/r)\ ] = (3q_a/2\pi k) \ln(D_{eq}/r) \quad \ldots\ldots(6)$$

For a balanced system, $V_{ab} + V_{ac} = (V_{an} - V_{bn}) + (V_{an} - V_{cn}) = 2V_{an} - (V_{bn} + V_{cn}) = 3V_{an}$. So from (6):

$$3V_{an} = (3q_a/2\pi k) \ln(D_{eq}/r) \Rightarrow V_{an} = (q_a/2\pi k) \ln(D_{eq}/r)$$

∴ Capacitance to neutral per phase: $C_n = q_a/V_{an} = 2\pi k / \ln(D_{eq}/r)$ F/m = 0.0242 / log(D_{eq}/r) μF/km (air, $k = \varepsilon_0$), with $D_{eq} = \sqrt[3]{(D_{12}\ D_{23}\ D_{31})}$

Conclusion: a transposed line with unsymmetrical spacing has the same capacitance as an equilateral line of spacing D_{eq}. Reference: K&N Ch-3, Art 3.6 and Example 3.4, p.106–107.

TERM TEST 2

Total marks: 10 · Time: 30 min

Question 1

Q1. A transmission line has a span of 300 m between level supports. The conductor has a cross-sectional area of 1.3 cm², weighs 1170 kg/km, and has a breaking stress of 4218 kg/cm². Calculate the sag for a safety factor of 5, allowing a wind pressure of 122 kg per square meter of projected area. What is the vertical sag? **[4]**

Solution:

Given: span l = 300 m; area A = 1.3 cm²; weight w = 1170 kg/km = 1.17 kg/m; breaking stress = 4218 kg/cm²; safety factor = 5; wind pressure = 122 kg/m².

Working tension:

$$T = (\text{Breaking stress} \times \text{Area}) / \text{Safety factor} = (4218 \times 1.3) / 5 = 1096.68 \text{ kg}$$

Diameter of conductor:

$$d = \sqrt{(4A/\pi)} = \sqrt{(4 \times 1.3 / \pi)} = 1.2866 \text{ cm} = 0.012866 \text{ m}$$

Wind force per metre length:

$$w_w = \text{Pressure} \times \text{Projected area} = 122 \times (d \times 1) = 122 \times 0.012866 \times 1 = 1.5696 \text{ kg/m}$$

Total (resultant) weight per metre:

$$w_t = \sqrt{(w^2 + w_w^2)} = \sqrt{(1.17^2 + 1.5696^2)} = \sqrt{(1.3689 + 2.4637)} = 1.9577 \text{ kg/m}$$

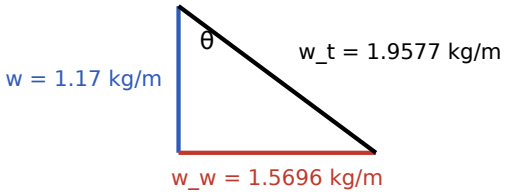


Fig. 4: Vector diagram of conductor weight, wind force and resultant (θ = angle of the plane of sag with the vertical)

Slant sag:

$$S = w_t l^2 / (8T) = (1.9577 \times 300^2) / (8 \times 1096.68) = 176193 / 8773.44 = 20.08 \text{ m}$$

Vertical sag:

$$\cos\theta = w / w_t = 1.17 / 1.9577 = 0.5976 \; (\theta = \tan^{-1}(w_w/w) = \tan^{-1}(1.3415) = 53.3°)$$

$$\text{Vertical sag} = S \cos\theta = 20.08 \times 0.5976 = 12.0 \text{ m}$$

∴ Slant sag S = 20.08 m; Vertical sag = 12.0 m

Reference: Mehta Ch-8, Example 8.19, p.191 (same problem with l = 200 m, A = 1.29 cm²); Art 8.16 (effect of wind and ice loading).

Question 2

Q2. A single-core cable is laid in the ground, the core diameter being 30 mm and the dielectric thickness 40 mm. The thermal resistivity of the dielectric is 5 thermal ohm-m, and the thermal resistance between the sheath and the ground surface is 0.45 thermal ohm per meter length of the cable. Neglecting dielectric and sheath losses, estimate the maximum permissible current loading if the temperature difference between the conductor and the ground surface is not to exceed 55 °C. The electrical resistance of the cable is 110 μΩ per metre length. **[3]**

Solution:

Given: core radius $r_1 = 30/2 = 15$ mm; outer radius of dielectric $r_2 = 15 + 40 = 55$ mm; thermal resistivity k = 5 thermal Ω-m; S_2 (sheath to ground) = 0.45 thermal Ω/m; t = 55 °C; R = 110 μΩ/m = 110×10^{-6} Ω/m; n = 1 (single core).

Thermal resistance of the dielectric per metre:

$$S_1 = (k / 2\pi) \ln(r_2 / r_1) = (5 / 2\pi) \ln(55 / 15) = 0.7958 \times 1.2993 = 1.034 \text{ thermal } \Omega/\text{m}$$

Total thermal resistance (conductor to ground surface):

$S = S_1 + S_2 = 1.034 + 0.45 = 1.484$ thermal Ω/m

Permissible current (heat generated = heat dissipated, $n\ I^2\ R = t / S$):

$I = \sqrt{(t / (n\ R\ S))} = \sqrt{(55 / (1 \times 110 \times 10^{-6} \times 1.484))}$

$= \sqrt{(55 / 1.6323 \times 10^{-4})} = \sqrt{336\ 944} = 580.5$ A

∴ Maximum permissible current loading, I ≈ 580 A

Reference: Mehta Ch-11, Example 11.20, p.292; Art 11.17–11.19 (thermal resistance, permissible current loading).

Question 3

Q3. Varley loop test is performed to locate an earth fault on a 40 km long cable. The resistance per km of the single conductor is 40 Ω. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 Ω. The fixed resistors have equal values. Calculate the distance of the fault from the test end. **[3]**

Solution:

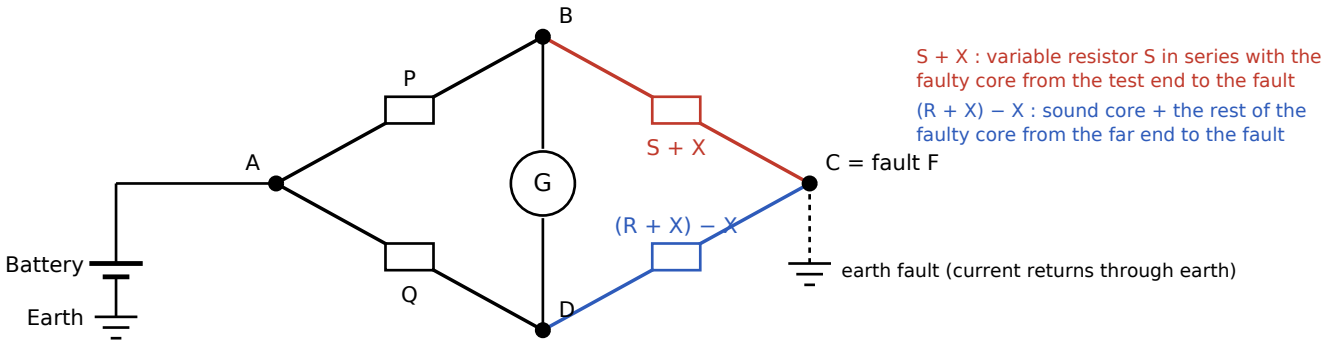


Fig. 5: Varley loop test (earth-fault position). X = resistance from the test end to the fault

Given: length = 40 km; resistance per km = 40 Ω; S = 200 Ω; P = Q.

Total loop resistance (faulty + healthy conductor):

$R + X = 2 \times 40\ \text{km} \times 40\ \Omega/\text{km} = 3200\ \Omega$

Balance condition of the Wheatstone bridge (earth-fault position):

$P / Q = (S + X) / [\ (R + X) - X\ ]$

Since $P = Q$: $S + X = (R + X) - X$

$\Rightarrow 200 + X = 3200 - X$

$\Rightarrow 2X = 3000 \Rightarrow X = 1500\ \Omega$

Distance of the fault:

Distance = X / (resistance per km) = 1500 / 40 = 37.5 km

∴ The fault is 37.5 km from the test end.

General formula (from the balance equation): $X = [P(R + X) - Q \cdot S] / (P + Q)$. Reference: Mehta Ch-11, Art 11.23 (Varley loop test, p.295) and Example 11.24, p.297.

TERM TEST 3

Total marks: 10 · Time: 30 min

Question 1

Q1. Using P-δ curve explain the phenomenon of a sudden change in mechanical input. **[4]**

Solution:

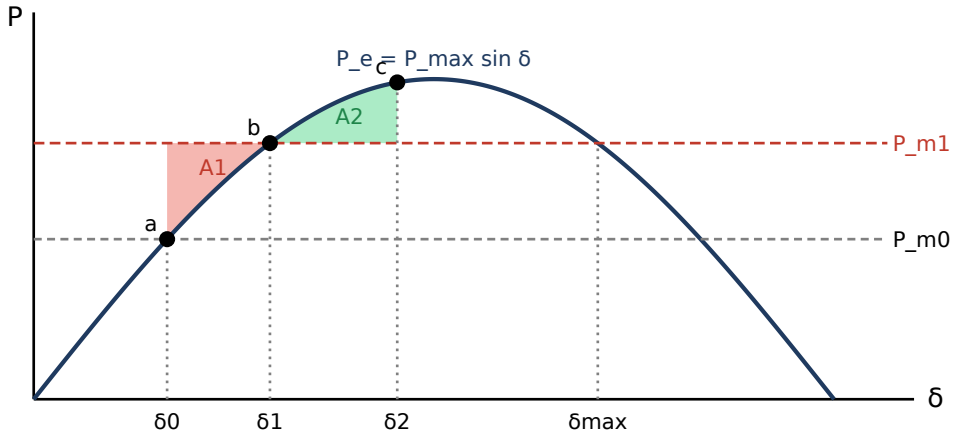


Fig. 6: Power-angle (P–δ) curve for a sudden increase in mechanical input from P_{m0} to P_{m1} (A1 = A2)

Consider a synchronous generator connected to an infinite bus. The electrical output is $P_e = (|E||V|/X) \sin \delta = P_{max} \sin \delta$, and the rotor obeys the swing equation $M\, d^2\delta/dt^2 = P_m - P_e$.

1. Initial state (point a): the machine runs at synchronous speed with input $P_{m0} = P_e$, at load angle δ_0.

2. Sudden increase: the mechanical input is suddenly raised to P_{m1}. Because of rotor inertia, δ cannot change instantly, so at δ_0 $P_{m1} > P_e$. The accelerating power $(P_{m1} - P_e)$ makes the rotor speed up above synchronous speed, and δ increases.

3. Point b (δ_1): here $P_e = P_{m1}$ (the new equilibrium), but the rotor is now running above synchronous speed, so δ keeps increasing past δ_1. The kinetic energy gained from δ_0 to δ_1 is the area **A1**.

4. Beyond δ_1: now $P_e > P_{m1}$, so the power is decelerating; the rotor slows down and gives back the stored energy (area **A2**). δ stops increasing at **δ_2 (point c)**, where the rotor speed is again synchronous, i.e. when **A2 = A1**.

5. Oscillation: the rotor then swings back and oscillates between δ_0 and δ_2 about δ_1; damping makes it settle at the new operating point b (δ_1). The system is **stable**.

Equal-area criterion:

$A1 = \int_{\delta 0}^{\delta 1} (P_{m1} - P_{max} \sin \delta)\, d\delta$ and $A2 = \int_{\delta 1}^{\delta 2} (P_{max} \sin \delta - P_{m1})\, d\delta$; stable if A2 = A1 can be reached

Stability limit: δ_2 can increase up to $\delta_{max} = \pi - \delta_1$. If A1 is larger than the whole area available above P_{m1} (between δ_1 and δ_{max}), δ goes beyond δ_{max}, P_{m1} becomes greater than P_e again, and the machine loses synchronism (unstable). So there is a maximum sudden increase in input that the system can tolerate, even though $P_{m1} < P_{max}$.

∴ A sudden input increase is stable only if the decelerating area A2 can become equal to the accelerating area A1 before δ reaches $\delta_{max} = \pi - \delta_1$.

Reference: K&N Ch-12, equal-area criterion: application to a sudden increase in power input; Lecture notes Lec 14–15.

Question 2

Q2. A 3-phase transmission line is being supported by three disc insulators. The potentials across top unit (i.e., near to the tower) and middle unit are 8 kV and 11 kV respectively. Calculate (i) the ratio of capacitance between pin and earth to the self-capacitance of each unit, (ii) the line voltage and (iii) string efficiency. **[6]**

Solution:

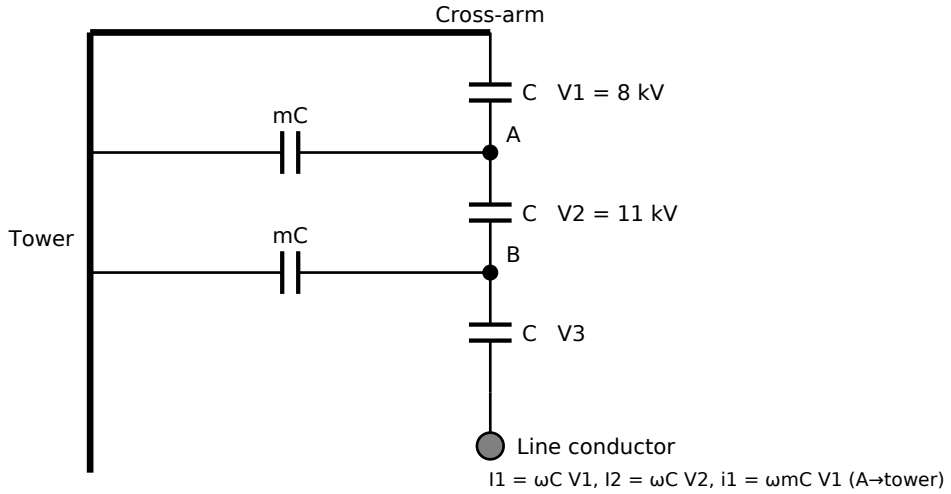


Fig. 7: Equivalent circuit of a 3-disc string. C = self-capacitance of each unit; mC = shunt (pin-to-earth) capacitance

Given: V_1 (top unit) = 8 kV, V_2 (middle unit) = 11 kV. Let m = (pin-to-earth capacitance)/(self-capacitance) = K.

(i) Ratio m (K): Applying Kirchhoff's current law at junction A:

$I_2 = I_1 + i_1$

$\omega C\, V_2 = \omega C\, V_1 + \omega mC\, V_1 \Rightarrow V_2 = V_1(1 + m)$

$11 = 8(1 + m) \Rightarrow 1 + m = 1.375 \Rightarrow m = 0.375$

Voltage across the bottom unit — KCL at junction B:

$I_3 = I_2 + i_2$

$\omega C\, V_3 = \omega C\, V_2 + \omega mC\, (V_1 + V_2) \Rightarrow V_3 = V_2 + m(V_1 + V_2)$

$V_3 = 11 + 0.375 \times (8 + 11) = 11 + 7.125 = 18.125$ kV

(ii) Line voltage:

Voltage across the string (phase voltage) $V = V_1 + V_2 + V_3 = 8 + 11 + 18.125 = 37.125$ kV

Line voltage $= \sqrt{3} \times V = \sqrt{3} \times 37.125 = 64.30$ kV

(iii) String efficiency:

η = Voltage across the string / (n × Voltage across the disc nearest the conductor)

$= 37.125 / (3 \times 18.125) \times 100\ \% = 37.125 / 54.375 \times 100\ \% = 68.28\ \%$

∴ (i) m = 0.375; (ii) Line voltage = 64.3 kV; (iii) String efficiency = 68.3 %

Reference: Mehta Ch-8, Art 8.6–8.7 (potential distribution and string efficiency, p.168); same type as Mehta Example 8.1–8.3. Also asked in Final 2024 (Session 2021-22) Q5(c).